THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. How many centimetres are there in 3 m? 1

(a) 300 **(b)** 290 **(c)** 1300 **(d)** 310

2. 3 chapatis are shared equally among 4 children. Each child gets (of a chapati): 1

(a) $\frac{1}{2}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{4}$ **(d)** $1\frac{1}{3}$

3. The number of pairs of equal opposite sides in a rectangle is: 1

(a) 3 **(b)** 2 **(c)** 5 **(d)** 6

4. The smallest angle of rotation of a figure is 45°. Its order of rotational symmetry is: 1

(a) 10 **(b)** 9 **(c)** 8 **(d)** 7

5. A person has a debt of ₹40 and then earns ₹90. The balance (₹) is: 1

(a) 50 **(b)** 60 **(c)** 52 **(d)** 48

6. The area of a triangle with base 14 cm and height 6 cm is (in cm^2): 1

(a) 48 **(b)** 28 **(c)** 56 **(d)** 42

7. $\frac{2}{3} + \frac{1}{4} =$ 1

(a) $\frac{5}{12}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{11}{12}$ **(d)** $\frac{3}{7}$

8. The perimeter of a rectangle with sides 11 cm and 4 cm is (cm): 1

(a) 30 **(b)** 41 **(c)** 34 **(d)** 19

9. A parallelogram (not a rectangle or rhombus) has how many lines of symmetry? 1

(a) 0 **(b)** 2 **(c)** 1 **(d)** 4

10. -9 + (-8) = 1

(a) -15 **(b)** -1 **(c)** -19 **(d)** -17

11. A right-angled triangle has legs 5 cm and 12 cm. Its area (in cm^2) is: 1

(a) 25 **(b)** 30 **(c)** 42 **(d)** 35

12. $\frac{5}{6} - \frac{1}{2} =$ 1

(a) $\frac{1}{3}$ **(b)** 1 **(c)** $1\frac{1}{3}$ **(d)** $\frac{1}{2}$

13. A circle has centre O and radius 5 cm. A point P is 8 cm from O. The point P lies: 1

(a) at the centre of the circle **(b)** outside the circle

(c) inside the circle **(d)** on the circle

14. The smallest angle through which a regular hexagon must be rotated to coincide with itself is (degrees): 1

(a) 60 **(b)** 90 **(c)** 50 **(d)** 70

15. -4 - (-9) = 1

(a) 5 **(b)** 3 **(c)** 7 **(d)** 23

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A plot is 30 m by 20 m. Find the cost of fencing it at ₹15 per metre. 2

17. Convert $2\frac{3}{5}$ to an improper fraction. 2

18. AB = 6 cm. Describe how to find the point equidistant from A and B on AB and its distance from A. 2

19. Write the mirror image of the word MAM in a vertical mirror (the reflection reads reversed) and say if the word looks the same. 2

20. Find $7 - 3$ and $3 - 7$. What do you notice? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the value: $5 - 9 + (-3)$. 3

22. A rectangle is 21 cm long and 10 cm wide. Find its perimeter and area. 3

23. Find the sum of $\frac{1}{2}$ and $\frac{2}{5}$ and compare the sum with 1. 3

24. Describe the construction of the perpendicular bisector of a line segment of length 7 cm. 3

OR

A regular polygon has rotational symmetry of angle 45°. How many sides does it have, and how many lines of symmetry? 3

25. Evaluate $-6 + (-4) + 15$. 3

OR

Find the area of a right-angled triangle with sides 5 cm, 12 cm and hypotenuse 13 cm, and its perimeter. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Letters of the word MATH: find which letters have vertical symmetry, which have horizontal symmetry and which have rotational symmetry of order 2. Is the word itself symmetric about a vertical mirror? 5

OR

A shopkeeper records profit as positive and loss as negative: Mon 120, Tue -80, Wed 50, Thu -30 (in ₹). Find the total, the best day and the worst day. 5

27. An L-shaped figure is made by joining a rectangle 12 cm by 5 cm and a rectangle 6 cm by 4 cm (they do not overlap). Find the total area of the figure and by how much the larger part exceeds the smaller part. 5

OR

Riya reads $\frac{2}{5}$ of a book on Monday and $\frac{1}{4}$ of it on Tuesday. What fraction has she read in two days? What fraction is left? On which day did she read more? 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND